

# Causal Representation Learning

## 1 Introduction

The primary goal of causal representation learning is to recover high-level latent causal variables from low-level observations. We consider a multi-domain setting, where one observes data from several domains that are generated from a common latent structure alongside domain-specific latent factors. Our research question is whether these shared and domain-specific latent variables can be formally identified from the observed distributions.

Our work builds on the framework introduced by Sturua et al. [1], in which the latent space decomposes into a shared component and domain-specific components. We adopt this structural assumption, but extend the setting by considering more expressive mixing functions. Specifically, while [1] restricts to linear mixing in an unpaired setting, we tackle the challenge of polynomial mixing functions, which allow for a richer nonlinear structure in the relationship between latent variables and observations. To achieve identifiability, we assume access to the joint distribution. Finally, we demonstrate how this identifiability framework naturally extends to a broader, generalized class of mixing functions.

## 2 Problem setup

**add scm explanation!!** We assume that the latent vector  $Z$  is generated by a structural causal model and that the observed variables arise through domain-specific polynomial mixing functions. In the two-domain case, the latent decomposition is

$$Z = (Z_{\mathcal{L}}, Z_{I_1}, Z_{I_2}),$$

where

$$Z_{\mathcal{L}} \in \mathbb{R}^p, \quad Z_{I_1} \in \mathbb{R}^{q_1}, \quad Z_{I_2} \in \mathbb{R}^{q_2}.$$

The variables in  $Z_{\mathcal{L}}$  are shared across the two domains, while  $Z_{I_1}$  and  $Z_{I_2}$  are domain-specific.

The observed variables are generated as

$$X^1 = G^1(Z_{\mathcal{L}}, Z_{I_1}), \quad X^2 = G^2(Z_{\mathcal{L}}, Z_{I_2}). \quad (1)$$

where  $G^1$  and  $G^2$  are called mixing functions.

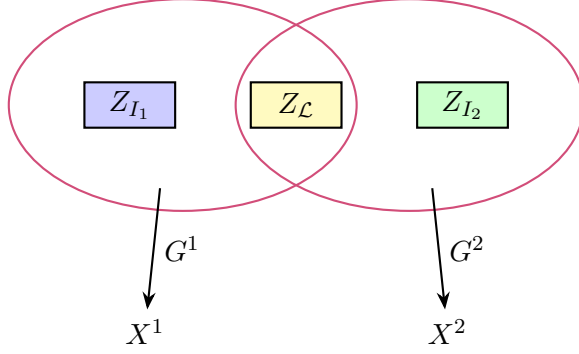


Figure 1: *Multi-domain causal graph  $\mathcal{G}$  with latent space decomposition. The shared space  $Z_{\mathcal{L}} \in \mathbb{R}^p$  and domain-specific spaces  $Z_{I_i} \in \mathbb{R}^{q_i}$  are identified by analyzing the rank of the polynomial mappings  $G^1$  and  $G^2$ .*

We assume access to the marginal distributions  $P_{X^1}$  and  $P_{X^2}$ , as well as to the joint distribution  $P_{X^1, X^2}$ . Initially, it is not known a priori whether a given variable  $Z_i$  is shared across domains or is domain-specific. Furthermore, the total number of shared latent variables is not fixed or known beforehand.

Let  $\phi : \mathbb{R}^{p+q_e} \rightarrow \mathbb{R}^m$  denote the vector of all monomials up to degree  $r$  in the components of  $(Z_{\mathcal{L}}, Z_{I_e})$ :

$$\phi(Z_{\mathcal{L}}, Z_{I_e}) = (1, Z_1, \dots, Z_{p+q_e}, Z_1^2, Z_1 Z_2, \dots, Z_{p+q_e}^r)^\top \quad (2)$$

where  $e \in \{1, 2\}$ . Then the mixing function takes the form:

$$X^e = G^e(Z_{\mathcal{L}}, Z_{I_e}) = C^e \cdot \phi(Z_{\mathcal{L}}, Z_{I_e}) \quad (3)$$

where  $C^e \in \mathbb{R}^{d_e \times m}$  is a coefficient matrix.

**Assumption 1.** *For each domain  $e \in \{1, 2\}$ , the map  $G^e$  is a full row rank polynomial map, and the relevant latent variables have support with non-empty interior.*

### 3 Identifiability of Latent Dimensionalities

A fundamental first step in our multi-domain framework is determining the dimensions of the shared and domain-specific latent spaces. Let  $R_1$ ,  $R_2$ , and  $R_{\text{tot}}$  denote the minimal latent dimensionalities required to generate the marginal distributions  $P_{X^1}$ ,  $P_{X^2}$ , and the joint distribution  $P_{X^1, X^2}$ , respectively. Leveraging the polynomial identifiability results established in [2], we show that these dimensionalities can be uniquely determined from observational data.

**Proposition 1.** *Under Assumption 1, the dimension of the shared latent space  $p$ , as well as the domain-specific dimensions  $q_1$  and  $q_2$ , are strictly identifiable from the marginal and joint observational distributions. Specifically, the shared dimension is given by:*

$$p = R_1 + R_2 - R_{\text{tot}}.$$

The full derivation of Proposition 1 is provided in Appendix B.

#### 3.1 Block-Triangular Representation

Even after recovering the value of  $p$ , the shared latent space itself is not immediately identified. Suppose that from the two marginal models we recover latent representations  $\hat{Z}^{(1)} \in \mathbb{R}^{p+q_1}$  and

$\hat{Z}^{(2)} \in \mathbb{R}^{p+q_2}$  corresponding to Domains 1 and 2. Then there exist invertible affine transformations relating these recovered variables to the true latent space:

$$\begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_1} \end{pmatrix} = \Lambda_1 \hat{Z}^{(1)} + b, \quad \begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_2} \end{pmatrix} = \Lambda_2 \hat{Z}^{(2)} + r, \quad (4)$$

where  $\Lambda_1, \Lambda_2$  are invertible matrices and  $b, r$  are vectors. Crucially, the true shared latent variable  $Z_{\mathcal{L}}$  should not depend on domain-specific coordinates. This structural requirement enforces a block-triangular form on the transformation matrices.

**Proposition 2.** *Assume the recovered representations can be partitioned into shared and private blocks  $\hat{Z}^{(1)} = (\hat{Z}_{\mathcal{L},1}^\top, \hat{Z}_{I_1}^\top)^\top$  and  $\hat{Z}^{(2)} = (\hat{Z}_{\mathcal{L},2}^\top, \hat{Z}_{I_2}^\top)^\top$ . Then, the affine mappings to the true latent space satisfy the block-triangular model:*

$$\begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_1} \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} \hat{Z}_{\mathcal{L},1} \\ \hat{Z}_{I_1} \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad (5)$$

and analogously for Domain 2.

The formal proof of Proposition 2 is deferred to Appendix C. This block-triangular structure is foundational for identifying the shared coordinates via conditional covariance in the subsequent analysis.

## 4 Identification via Conditional Covariance

We now address the harder problem: once the shared dimension  $p$  is known, how can we identify which recovered coordinates correspond to the shared variables?

Assume that there exist permutation matrices  $P_1$  and  $P_2$  such that the recovered latent coordinates can be reordered into shared and private blocks:

$$P_1 \hat{Z}^{(1)} = \begin{pmatrix} \hat{Z}_{\mathcal{L},1} \\ \hat{Z}_{I_1} \end{pmatrix}, \quad P_2 \hat{Z}^{(2)} = \begin{pmatrix} \hat{Z}_{\mathcal{L},2} \\ \hat{Z}_{I_2} \end{pmatrix}.$$

As established in Section 3.1, these reordered representations satisfy the block-triangular model:

$$\begin{pmatrix} \hat{Z}_{\mathcal{L},1} \\ \hat{Z}_{I_1} \end{pmatrix} = \begin{pmatrix} A_1 & 0 \\ C_1 & D_1 \end{pmatrix} \begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_1} \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad (6)$$

$$\begin{pmatrix} \hat{Z}_{\mathcal{L},2} \\ \hat{Z}_{I_2} \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ G_1 & H_1 \end{pmatrix} \begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_2} \end{pmatrix} + \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}. \quad (7)$$

We additionally assume conditional independence of the private latent variables across domains.

**Assumption 2.** *The domain-specific latent variables satisfy  $Z_{I_1} \perp Z_{I_2} \mid Z_{\mathcal{L}}$ .*

We now prove that the shared coordinates can be exactly identified from the diagonal of the conditional covariance matrix between the recovered representations.

**Theorem 1.** *Given the recovered representations satisfy the block-triangular model (6)–(7), with  $A_1, E_1$  invertible, and assuming Assumption 2 holds. Let  $\Sigma_{1|2} := \mathbb{E} [\text{Cov}(\hat{Z}^{(1)} \mid \hat{Z}^{(2)})]$ . Then the indices corresponding to the shared coordinates in  $\hat{Z}^{(1)}$  are exactly the zero diagonal entries of  $\Sigma_{1|2}$ , while the indices corresponding to the private coordinates are exactly the strictly positive diagonal entries. Analogously, the shared and private coordinates in  $\hat{Z}^{(2)}$  are identified from  $\Sigma_{2|1}$ .*

*Proof Sketch.* The full proof is deferred to Appendix D.1. The intuition relies on the block-triangular structure: because  $E_1$  is invertible, conditioning on  $\hat{Z}^{(2)}$  perfectly determines the shared true latent  $Z_{\mathcal{L}}$ . This removes all variance from the shared recovered block  $\hat{Z}_{\mathcal{L},1}$ , forcing its conditional covariance to zero. In contrast, the conditional independence (Assumption 2) ensures the private block  $\hat{Z}_{I_1}$  retains strictly positive variance.  $\square$

Theorem 1 allows us to formally define the shared and private index sets in Domain 1 as:

$$\mathcal{I}_L^{(1)} = \{i \in [p + q_1] : (\Sigma_{1|2})_{ii} = 0\}, \quad \mathcal{I}_{I_1}^{(1)} = \{i \in [p + q_1] : (\Sigma_{1|2})_{ii} > 0\}.$$

By symmetry, we obtain the analogous index sets  $\mathcal{I}_L^{(2)}$  and  $\mathcal{I}_{I_2}^{(2)}$  for Domain 2 using  $\Sigma_{2|1}$ .

Having identified which coordinates correspond to the private variables, a natural follow-up question is how well we can recover the true private latent variables  $Z_{I_1}$  from the recovered representation  $\hat{Z}_{I_1}$ . Specifically, what is the best distribution  $p$  for  $Z_{I_1}$  such that it perfectly explains what we observe?

Theorem 2 formalizes this by defining the optimal distribution  $p^*$  conditional on a fixed shared representation  $\hat{Z}_{\mathcal{L}} = z$ , achieving a perfect correlation of 1. However, this reveals a fundamental limit of identification from observational data alone (illustrated in Figure 2): the optimal distribution is identifiable only up to an invertible affine transformation.

**Theorem 2.** *Under Assumptions 1 and 2, if  $\text{Cov}(Z_{I_1} \mid Z_{\mathcal{L}}) \succ 0$  and given that the shared coordinates have been identified via Theorem 1, fix  $\hat{Z}_{\mathcal{L}} = z$ . Define  $c(z) := C_1 A_1^{-1} z - C_1 A_1^{-1} b_1 + b_2$ . Then the optimal distribution*

$$p^* := \mathcal{L}(D_1^{-1}(\hat{Z}_{I_1} - c(z)) \mid \hat{Z}_{\mathcal{L}} = z) \quad (8)$$

*achieves  $\text{corr}(Z_{I_1}, \hat{Z}_{I_1} \mid \hat{Z}_{\mathcal{L}} = z) = 1$ , and is unique up to an invertible affine transformation.*

The full proof is provided in Appendix D.2.

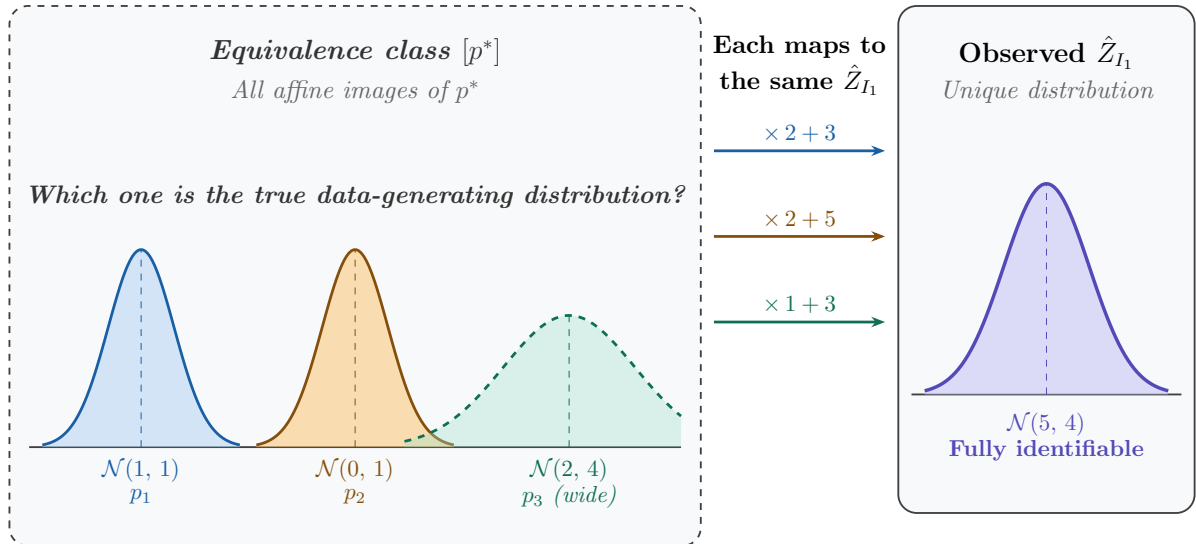


Figure 2: **Fundamental limit of identification from observational data.** The observed distribution of the recovered private variables ( $\hat{Z}_{I_1}$ ) pins down the affine equivalence class  $[p^*]$ . Any distribution within this class can be mapped via an invertible affine transformation to produce the exact same observable outcome.

Simulations/Experiments...

## 5 Simulations

We evaluate our method in two settings: a *Gaussian* setting that matches the theoretical assumptions of the polynomial mixing framework, and a *general* setting in which during the formation each latent node receives random noise. The general setting tests robustness beyond the identifiability assumptions and covers four noise families: Laplace, Uniform, Exponential, and Gaussian mixture.

**Data generating process.** In both settings we fix  $p = 3$ ,  $q_1 = q_2 = 3$ , so the total latent dimension is  $h = 9$ . The latent variables are generated from an SCM on a randomly sampled Erdős–Rényi DAG with edge probability 0.4 and edge weights drawn uniformly from  $[-1, -0.25] \cup [0.25, 1]$ . Observed variables are generated via full-rank polynomial mixing maps of degree  $r = 2$  with  $d_{\text{obs}} = 10$ :

$$X_e = G_e(\varphi_r(Z_L, Z_{I_e})), \quad e \in \{1, 2\}.$$

Full details of the noise distributions and algorithmic procedures are given in Appendix F.

**Results.** Figure ?? shows that in the Gaussian setting the MAE decreases toward zero and the Jaccard similarity approaches 1 as  $n$  grows, confirming the consistency of our estimator. Figure ?? shows that the flow-based extension retains this behaviour across all four non-Gaussian noise families, demonstrating that identifiability extends beyond the Gaussian case.

## 6 Generalization to Other Mixing Functions

While our primary focus thus far has been resolving the identifiability of shared latent spaces under polynomial mixing, our framework naturally extends to a broader class of nonlinear mixing functions. We challenge the requirement of linear mixing by defining a general map that relates the latent space to observable data via a set of basis components.

**Definition 1.** Let  $Z \in \mathbb{R}^p$  denote the latent variables. The observable data  $X \in \mathbb{R}^n$  is generated by a mixing function of the form:

$$X = C\Phi(Z) + b, \tag{9}$$

where  $\Phi(Z) = (\phi_1(Z), \dots, \phi_m(Z))^\top$  is a vector of continuous basis components (e.g., monomials or exponentials),  $C \in \mathbb{R}^{n \times m}$  is a mixing coefficient matrix with full column rank, and  $b \in \mathbb{R}^n$  is a bias vector.

**Assumption 3.** We assume the following regularity conditions:

- (i) The interior of the support of the latent distribution  $\mathbb{P}_Z$  is a non-empty subset of  $\mathbb{R}^p$ .
- (ii) The basis components  $\{1, \phi_1, \dots, \phi_m\}$  of the mixing function are linearly independent on any non-empty open set.

Under these conditions, the algebraic structure of the basis functions imposes sufficient constraints to identify the basis representations up to an affine transformation.

**Theorem 3.** Suppose two causal representations  $(C, b, \Phi, Z)$  and  $(\tilde{C}, \tilde{b}, \tilde{\Phi}, \tilde{Z})$  satisfy Assumption 3 and both generate the exact same distribution of observable data  $\mathbb{P}_X$ . Then, the basis representations are identifiable up to an invertible affine transformation:

$$\Phi(Z) = M\tilde{\Phi}(\tilde{Z}) + v, \tag{10}$$

where  $M \in \mathbb{R}^{m \times m}$  is an invertible matrix and  $v \in \mathbb{R}^m$ .

The proof of Theorem 3 is provided in Appendix E.1.

Consequently, for specific families of mixing functions where linear dependencies in the basis space strictly imply affine dependencies in the latent space, Theorem 3 guarantees that the latent space  $Z$  is identifiable up to an invertible affine transformation  $\tilde{Z} = \Lambda Z + r$ . Polynomial mixing, as analyzed in previous sections, is one such family. We now demonstrate that exponential families also satisfy this property.

**Exponential Identifiability.** Following [2], we consider observations generated by a weighted sum of exponentials of the latent variables:

$$\Phi(Z) = (\exp(k_1^\top Z), \dots, \exp(k_m^\top Z))^\top, \quad (11)$$

where  $k_i \in \mathbb{R}^p$  are fixed direction vectors. Each component captures the latent variable  $Z$  from a different direction. Crucially, exponentials with distinct exponent vectors are linearly independent on any non-empty open set, satisfying Assumption 3(ii).

**Lemma 1.** *Under Assumption 3, if the mixing function consists of exponential basis components, the dimension  $p$  of  $Z$  is identifiable, and  $Z$  is identifiable up to an invertible affine transformation  $\Lambda Z + c$  for some non-singular  $\Lambda \in \mathbb{R}^{p \times p}$  and  $c \in \mathbb{R}^p$ . With observational data alone, this affine ambiguity cannot be reduced further.*

The proof of Lemma 1, which applies the result of Theorem 3 to the exponential basis, is detailed in Appendix E.2.

## A Polynomial Identifiability

In this section, we restate a foundational identifiability result for polynomial mixing functions established by Zhang et al. [2]. This result forms the theoretical basis for our dimensionality recovery in Proposition 1, as it guarantees that the minimal latent dimensionality of a full row rank polynomial map can be uniquely determined from its marginal observational distribution.

**Lemma 2** (Zhang et al. [2]). *Under Assumption 1, we can identify the dimension  $p$  of  $U$  as well as its linear transformation  $U\Lambda + b$  for some non-singular matrix  $\Lambda$  and vector  $b$ . In fact, with observational data, we can only identify  $U$  up to such linear transformations.*

## B Derivation of Latent Dimensionalities

Here is the proof of the Proposition 1.

*Proof. Domain 1 and 2.* The latent variables entering the mixing function for Domain 1 are  $(Z_{\mathcal{L}}, Z_{I_1})$ , yielding a total dimension of  $R_1 := p + q_1$ . Similarly, for Domain 2, the total latent dimension is  $R_2 := p + q_2$ . By the polynomial identifiability result presented in [2], the dimensions  $R_1$  and  $R_2$  are identifiable from the marginal observational distributions  $P_{X^1}$  and  $P_{X^2}$ , respectively. Under the full row rank assumption, these are precisely the minimal latent dimensionalities needed to generate the two marginal distributions.

**Joint latent dimension.** Now consider the joint distribution  $P_{X^1, X^2}$ . Since both domains are generated from the combined latent collection  $(Z_{\mathcal{L}}, Z_{I_1}, Z_{I_2})$ , the total latent dimension in the joint model is:

$$R_{\text{tot}} := p + q_1 + q_2. \quad (12)$$

By applying the same identifiability principle to the joint observational law, this combined quantity  $R_{\text{tot}}$  is also identifiable.

**Recovering the dimensions.** We can now isolate the shared dimension by evaluating the overlap between the dimensionalities of the individual domains and the joint model:

$$R_1 + R_2 - R_{\text{tot}} = (p + q_1) + (p + q_2) - (p + q_1 + q_2) = p. \quad (13)$$

Analogously, the dimensions of the domain-specific spaces are uniquely recovered as  $q_1 = R_{\text{tot}} - R_2$  and  $q_2 = R_{\text{tot}} - R_1$ .  $\square$

## C Proof of Block-Triangular Representation

Here we present the proof of Proposition 2.

*Proof.* Partition the affine maps as:

$$\Lambda_1 = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad \Lambda_2 = \begin{pmatrix} E & F \\ G & H \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad r = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}. \quad (14)$$

Expanding the transformation for Domain 1 yields:

$$\begin{pmatrix} Z_{\mathcal{L}} \\ Z_{I_1} \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} \hat{Z}_{\mathcal{L},1} \\ \hat{Z}_{I_1} \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}. \quad (15)$$

Looking strictly at the first block row, we obtain:

$$Z_{\mathcal{L}} = A\hat{Z}_{\mathcal{L},1} + B\hat{Z}_{I_1} + b_1. \quad (16)$$

By definition, the true shared latent variable  $Z_{\mathcal{L}}$  must not depend on the domain-specific recovered coordinates. To formalize this, fix the shared recovered coordinates to a constant  $\hat{Z}_{\mathcal{L},1} = s$ , and consider two distinct attainable values  $d, d'$  of the private recovered coordinates  $\hat{Z}_{I_1}$ . Equation (16) gives:

$$Z_{\mathcal{L}} = As + Bd + b_1 \quad \text{and} \quad Z_{\mathcal{L}} = As + Bd' + b_1. \quad (17)$$

Subtracting the two identities yields:

$$B(d - d') = 0. \quad (18)$$

Since this equality must hold for all such pairs of attainable values, it necessarily follows that  $B = 0$ .

Applying the identical argument to the affine transformation for Domain 2, we examine its first block row:

$$Z_{\mathcal{L}} = E\hat{Z}_{\mathcal{L},2} + F\hat{Z}_{I_2} + r_1. \quad (19)$$

Fixing  $\hat{Z}_{\mathcal{L},2}$  and varying  $\hat{Z}_{I_2}$  analogously forces  $F = 0$ . Thus, the off-diagonal blocks vanish, and both  $\Lambda_1$  and  $\Lambda_2$  are block-triangular, concluding the proof.

The similar argument can be applied to the lower-left block  $C$ . Indeed, the second block row gives

$$Z_{I_1} = C\hat{Z}_{\mathcal{L},1} + D\hat{Z}_{I_1} + b_2.$$

Varying  $\hat{Z}_{\mathcal{L},1}$  while keeping  $\hat{Z}_{I_1}$  fixed may change  $Z_{I_1}$ , we get  $C = 0$ . Analogously, we obtain  $G = 0$ .  $\square$

## D Proofs for Identification via Conditional Covariance

### D.1 Proof of Theorem 1

We analyze the reordered latent representations  $\hat{Z}^{(1)} := (\hat{Z}_{\mathcal{L},1}^\top, \hat{Z}_{I_1}^\top)^\top$  and  $\hat{Z}^{(2)} := (\hat{Z}_{\mathcal{L},2}^\top, \hat{Z}_{I_2}^\top)^\top$ .

**Step 1.** From the block-triangular model, the shared block of Domain 2 is  $\hat{Z}_{\mathcal{L},2} = E_1 Z_{\mathcal{L}} + r_1$ . Since  $E_1$  is invertible, conditioning on  $\hat{Z}^{(2)}$  reveals  $Z_{\mathcal{L}}$  exactly:

$$Z_{\mathcal{L}} = E_1^{-1}(\hat{Z}_{\mathcal{L},2} - r_1).$$

Now for Domain 1,  $\hat{Z}_{\mathcal{L},1} = A_1 Z_{\mathcal{L}} + b_1$ . Thus, once  $\hat{Z}^{(2)}$  is given,  $\hat{Z}_{\mathcal{L},1}$  is completely determined. Therefore,  $\mathbb{E}[\hat{Z}_{\mathcal{L},1} | \hat{Z}^{(2)}] = \hat{Z}_{\mathcal{L},1}$ , which implies:

$$\text{Cov}(\hat{Z}_{\mathcal{L},1} | \hat{Z}^{(2)}) = 0. \quad (20)$$

**Step 2.** For the private block of Domain 1,  $\hat{Z}_{I_1} = C_1 Z_{\mathcal{L}} + D_1 Z_{I_1} + b_2$ . Conditioning on  $\hat{Z}^{(2)}$  reveals  $Z_{\mathcal{L}}$ , so the term  $C_1 Z_{\mathcal{L}} + b_2$  becomes deterministic.

Using the conditional independence assumption  $Z_{I_1} \perp Z_{I_2} | Z_{\mathcal{L}}$ , and the fact that  $\hat{Z}^{(2)}$  depends only on  $(Z_{\mathcal{L}}, Z_{I_2})$ , we obtain  $\mathbb{E}[Z_{I_1} | \hat{Z}^{(2)}] = \mathbb{E}[Z_{I_1} | Z_{\mathcal{L}}]$ . Hence,

$$\mathbb{E}[\hat{Z}_{I_1} | \hat{Z}^{(2)}] = C_1 Z_{\mathcal{L}} + D_1 \mathbb{E}[Z_{I_1} | Z_{\mathcal{L}}] + b_2.$$

Subtracting this from  $\hat{Z}_{I_1}$  yields  $\hat{Z}_{I_1} - \mathbb{E}[\hat{Z}_{I_1} | \hat{Z}^{(2)}] = D_1 (Z_{I_1} - \mathbb{E}[Z_{I_1} | Z_{\mathcal{L}}])$ . Therefore, the conditional covariance is:

$$\text{Cov}(\hat{Z}_{I_1} | \hat{Z}^{(2)}) = D_1 \text{Cov}(Z_{I_1} | Z_{\mathcal{L}}) D_1^\top. \quad (21)$$

If  $\text{Cov}(Z_{I_1} | Z_{\mathcal{L}}) \succ 0$  and  $D_1$  has full column rank, then this matrix has strictly positive diagonal entries.

**Proposition 3.** *Under Assumption 1 and the structural causal model (SCM), the conditional covariance matrix  $\text{Cov}(Z_{I_1} | Z_{\mathcal{L}})$  is positive definite.*

*Proof.* By the SCM definition, the vector of domain-specific latent variables  $Z_{I_1}$  is generated by:

$$Z_{I_1} = W_{\mathcal{L}} Z_{\mathcal{L}} + W_{I_1} Z_{I_1} + \epsilon_{I_1} \quad (22)$$

where  $W_{\mathcal{L}} \in \mathbb{R}^{q_1 \times p}$  and  $W_{I_1} \in \mathbb{R}^{q_1 \times q_1}$  represent the causal weights from shared and private parents, respectively, and  $\epsilon_{I_1}$  is a vector of independent noise terms.

Since the causal graph  $\mathcal{G}$  is a DAG, the matrix  $(I - W_{I_1})$  is lower triangular and thus invertible. We can solve for  $Z_{I_1}$  as:

$$Z_{I_1} = (I - W_{I_1})^{-1} W_{\mathcal{L}} Z_{\mathcal{L}} + (I - W_{I_1})^{-1} \epsilon_{I_1} \quad (23)$$

When conditioning on  $Z_{\mathcal{L}}$ , the term  $(I - W_{I_1})^{-1} W_{\mathcal{L}} Z_{\mathcal{L}}$  becomes deterministic. The conditional distribution of the domain-specific block is:

$$Z_{I_1} | Z_{\mathcal{L}} = (I - W_{I_1})^{-1} \epsilon_{I_1} + \text{constant} \quad (24)$$

The conditional covariance is then given by:

$$\text{Cov}(Z_{I_1} | Z_{\mathcal{L}}) = (I - W_{I_1})^{-1} \text{Cov}(\epsilon_{I_1}) ((I - W_{I_1})^{-1})^\top \quad (25)$$

By Assumption A, the latent variables have support with a non-empty interior. This implies that each noise component  $\epsilon_j$  has strictly positive variance,  $\sigma_j^2 > 0$ . Consequently,  $\text{Cov}(\epsilon_{I_1})$  is a diagonal matrix with strictly positive diagonal entries, satisfying  $\text{Cov}(\epsilon_{I_1}) \succ 0$ .

Since  $(I - W_{I_1})^{-1}$  is non-singular,  $(I - W_{I_1})^{-1} \text{Cov}(\epsilon_{I_1}) ((I - W_{I_1})^{-1})^\top$  preserves positive definiteness. Thus,  $\text{Cov}(Z_{I_1} | Z_{\mathcal{L}}) \succ 0$ , ensuring its diagonal entries are strictly positive.  $\square$



**Step 3.** Combining the previous steps, the conditional covariance of the reordered representation is:

$$\hat{\Sigma}_{1|2} = \begin{pmatrix} 0_{p \times p} & 0 \\ 0 & D_1 \text{Cov}(Z_{I_1} | Z_{\mathcal{L}}) D_1^\top \end{pmatrix}.$$

Because  $\hat{Z}_{\mathcal{L},1}$  is deterministic given  $\hat{Z}^{(2)}$ , it cannot co-vary with the remaining uncertainty in the private representation  $\hat{Z}_{I_1}$ . Thus, the off-diagonal cross-covariance blocks vanish. The first  $p$  diagonal entries are exactly zero, and the remaining  $q_1$  entries are strictly positive.

**Step 4.** The observed representation  $\hat{Z}^{(1)}$  is related to the reordered one by the permutation matrix  $P_1$ , meaning  $\Sigma_{1|2} = P_1^\top \hat{\Sigma}_{1|2} P_1$ . Because permutation preserves the diagonal values, the zero diagonal entries of  $\Sigma_{1|2}$  exactly identify the locations of the shared coordinates in the original ordering.  $\square$

## D.2 Proof of Theorem 2

We know that  $\hat{Z}_{I_1} = C_1 A_1^{-1}(\hat{Z}_{\mathcal{L}} - b_1) + D_1 Z_{I_1} + b_2$ . Therefore, it holds that:

$$\hat{Z}_{I_1} | \hat{Z}_{\mathcal{L}} = z \stackrel{\text{a.s.}}{=} D_1 Z_{I_1} + c(z), \quad (26)$$

where  $c(z)$  is deterministic given  $z$ . Since  $Z_{I_1}$  and  $\hat{Z}_{I_1}$  are jointly defined on the same probability space, (26) is an exact algebraic identity, which immediately implies  $\text{corr}(Z_{I_1}, \hat{Z}_{I_1} | \hat{Z}_{\mathcal{L}} = z) = 1$ .

Since  $D_1$  is invertible, (26) gives  $Z_{I_1} = T_z(\hat{Z}_{I_1})$  almost surely. For any measurable set  $B \subseteq \mathbb{R}^{q_1}$ :

$$\begin{aligned} p^*(B) &= P(Z_{I_1} \in B) = P(\hat{Z}_{I_1} \in D_1 B + c(z) | \hat{Z}_{\mathcal{L}} = z) \\ &= \mathcal{L}(\hat{Z}_{I_1} | \hat{Z}_{\mathcal{L}} = z)(D_1 B + c(z)). \end{aligned} \quad (27)$$

For uniqueness, suppose  $p'$  also achieves a correlation of 1. Then there exist  $A', b'$  such that  $A Z_{I_1} + b \stackrel{d}{=} \hat{Z}_{I_1} | \hat{Z}_{\mathcal{L}} = z \stackrel{d}{=} A' Z'_{I_1} + b'$ , which gives:

$$Z'_{I_1} \stackrel{d}{=} (A')^{-1} A Z_{I_1} + (A')^{-1}(b - b'). \quad (28)$$

Thus,  $p'$  is an affine image of  $p^*$ . Finally, if  $f(Z_{I_1})$  also achieves a correlation of 1 for some measurable  $f$ , combining the two almost sure identities yields  $f(Z_{I_1}) \stackrel{\text{a.s.}}{=} \tilde{A}^{-1} D_1 Z_{I_1} + \tilde{A}^{-1}(c(z) - \tilde{b})$ , which is affine almost surely since  $Z_{I_1}$  has support with a non-empty interior. Thus, the equivalence class  $[p^*]$  is uniquely identified.  $\square$

## E Proofs for Generalization to Other Mixing Functions

### E.1 Proof of Theorem 3

Suppose the two representations generate the same observable data distribution. Because the mappings are deterministic, they must produce the same observable data  $X$  almost surely. Therefore, we have:

$$C\Phi(Z) + b = \tilde{C}\tilde{\Phi}(\tilde{Z}) + \tilde{b}. \quad (29)$$

Because  $C$  and  $\tilde{C}$  have full column rank, there exist left-inverses  $A$  and  $\tilde{A}$  such that  $AC = I_m$  and  $\tilde{A}\tilde{C} = I_{\tilde{m}}$ . Applying the left-inverse  $A$  to both sides of the equation yields:

$$\Phi(Z) = A\tilde{C}\tilde{\Phi}(\tilde{Z}) + A(\tilde{b} - b). \quad (30)$$

Let us define  $M := A\tilde{C}$  and  $v := A(\tilde{b} - b)$ . This gives us the mapping  $\Phi(Z) = M\tilde{\Phi}(\tilde{Z}) + v$ .

To prove that  $M$  is invertible, we apply the symmetric argument using the left-inverse  $\tilde{A}$ , which yields:

$$\tilde{\Phi}(\tilde{Z}) = \tilde{A}C\Phi(Z) + \tilde{A}(b - \tilde{b}) := \tilde{M}\Phi(Z) + \tilde{v}. \quad (31)$$

Substituting this back into our expression for  $\Phi(Z)$  gives us:

$$\Phi(Z) = M(\tilde{M}\Phi(Z) + \tilde{v}) + v = (M\tilde{M})\Phi(Z) + (M\tilde{v} + v). \quad (32)$$

By Assumption 3, the basis components of  $\Phi(Z)$  are linearly independent on the non-empty open support of  $Z$ . For the above equation to hold for all  $Z$  in the support, the coefficients must match the identity exactly. Therefore, we must have:

$$M\tilde{M} = I_m \quad \text{and} \quad M\tilde{v} + v = 0. \quad (33)$$

This implies that  $M$  is an invertible matrix (with inverse  $\tilde{M}$ ), proving that the basis representations are equivalent up to an invertible affine transformation.  $\square$

## E.2 Proof of Lemma 1

We follow the proof of [2], adapted to our setting. Suppose a second representation  $X = \hat{C}\hat{\Phi}(\hat{Z}) + \hat{b}$  generates the same distribution  $\mathbb{P}_X$ . From Theorem 3, applying the left-inverses  $A$  and  $\hat{A}$  to  $X$  and substituting one representation into the other yields the affine relationship:

$$\Phi(Z) = M\hat{\Phi}(\hat{Z}) + v, \quad \hat{\Phi}(\hat{Z}) = \hat{M}\Phi(Z) + \hat{v}, \quad (34)$$

where  $M := A\hat{C}$ ,  $\hat{M} := \hat{A}C$ ,  $v := A(\hat{b} - b)$ , and  $\hat{v} := \hat{A}(b - \hat{b})$ . Looking specifically at the  $i$ -th coordinate:

$$\exp(k_i^\top Z) = \sum_{j=1}^{\hat{m}} M_{ij} \exp(\hat{k}_j^\top \hat{Z}) + v_i. \quad (35)$$

Since finite families of exponentials with distinct exponent vectors are linearly independent on any non-empty open set, we must have  $v_i = 0$ , and all but one  $M_{ij}$  must vanish. Therefore, up to a permutation  $\pi$  of  $[\hat{m}]$ , it must hold that:

$$\exp(k_i^\top Z) = \alpha_i \exp(\hat{k}_{\pi(i)}^\top \hat{Z}), \quad \alpha_i \neq 0. \quad (36)$$

Taking the natural logarithm and stacking over all coordinates  $i$  gives:

$$KZ = \hat{K}_\pi \hat{Z} + \beta, \quad (37)$$

where  $\beta_i = \log \alpha_i$ . By assumption, the direction matrix  $K$  has full column rank. Multiplying by the pseudo-inverse  $K^\dagger$  and applying the symmetric argument forces the latent dimensions to match ( $p = \hat{p}$ ) and yields:

$$\hat{Z} = \Lambda Z + c, \quad (38)$$

for a non-singular transformation matrix  $\Lambda \in \mathbb{R}^{p \times p}$  and vector  $c \in \mathbb{R}^p$ .  $\square$

## F Simulation Details

### F.1 Noise Distributions

All noise distributions are parameterised to have zero mean and unit variance.

- **Gaussian:**  $\epsilon_j \sim \mathcal{N}(0, 1)$ .
- **Laplace:**  $\epsilon_j \sim \text{Laplace}(0, 1/\sqrt{2})$ , so  $\text{Var}(\epsilon_j) = 1$ .

- **Uniform:**  $\epsilon_j \sim \text{Uniform}(-\sqrt{3}, \sqrt{3})$ , so  $\text{Var}(\epsilon_j) = 1$ .
- **Exponential:**  $\epsilon_j \sim \text{Exp}(1) - 1$ , so  $\mathbb{E}[\epsilon_j] = 0$  and  $\text{Var}(\epsilon_j) = 1$ .
- **Gaussian mixture:**  $\epsilon_j \sim \frac{1}{2}\mathcal{N}(-1, 1) + \frac{1}{2}\mathcal{N}(1, 1)$ , so  $\mathbb{E}[\epsilon_j] = 0$  and  $\text{Var}(\epsilon_j) = 2$ .

Each node in the SCM independently draws its noise from one of the above families. In the *general* experiments, we run one trial per noise type and report results separately.

## F.2 Algorithms

### F.2.1 Gaussian Setting

Algorithm 1 recovers the latent dimension under the Gaussian assumption using a Cholesky-parameterised latent model. Algorithm 2 describes the full simulation pipeline.

---

#### Algorithm 1 Greedy Latent Dimension Recovery (Gaussian)

---

```

1: for  $\hat{p} = 1, 2, 3, \dots, p_{\max}$  do
2:   Initialize  $\mu \leftarrow 0$ ,  $L \leftarrow I_{\hat{p}}$ ,  $G \leftarrow$  small random
3:   for  $t = 1, \dots, T$  do
4:     Sample  $\epsilon \in \mathbb{R}^{n \times \hat{p}}$ , rows i.i.d.  $\mathcal{N}(0, I_{\hat{p}})$ 
5:      $\hat{Z} \leftarrow \mu + \epsilon \text{tril}(L)^\top$ 
6:      $\hat{X} \leftarrow \varphi(\hat{Z})G^\top$ , then standardize  $\hat{X}$ 
7:      $\mathcal{L} \leftarrow \text{MMD}^2(\hat{X}, X)$ 
8:     Update  $\mu$ ,  $L$ ,  $G$  via Adam
9:   end for
10:  if  $\mathcal{L} < \tau$  and  $\text{rank}(G) = n_\varphi$  then
11:    return  $\hat{p}$ 
12:  end if
13: end for

```

---



---

#### Algorithm 2 Simulation Pipeline (Gaussian)

---

**Require:**  $p, q_1, q_2, n, r, d_{\text{obs}}$

**Ensure:**  $\hat{p}, \hat{R}$

- 1: Sample weighted DAG on  $h = p + q_1 + q_2$  nodes; partition  $Z = (Z_L, Z_{I_1}, Z_{I_2})$  with Gaussian noise
  - 2: Simulate  $n$  samples from SCM
  - 3: Generate  $X_1 = G_1(\varphi_r(Z_L, Z_{I_1}))$  and  $X_2 = G_2(\varphi_r(Z_L, Z_{I_2}))$
  - 4: Form  $X_{\text{joint}} = (X_1, X_2)$
  - 5:  $\hat{d}_1 \leftarrow \text{Algorithm 1}(X_1)$ ;  $\hat{d}_2 \leftarrow \text{Algorithm 1}(X_2)$ ;  $\hat{d}_{\text{joint}} \leftarrow \text{Algorithm 1}(X_{\text{joint}})$
  - 6:  $\hat{p} \leftarrow \hat{d}_1 + \hat{d}_2 - \hat{d}_{\text{joint}}$  (*Proposition 1*)
  - 7: Compute  $\text{diag}(\hat{\Sigma}_{1|2})$  on  $X_1, X_2$  via Schur complement
  - 8:  $\hat{R} \leftarrow$  indices of  $\hat{p}$  smallest entries of  $\text{diag}(\hat{\Sigma}_{1|2})$
  - 9: **return**  $\hat{p}, \hat{R}$
- 

### F.2.2 General Setting

In the general setting, the Gaussian latent model  $\hat{Z} = \mu + \epsilon \text{tril}(L)^\top$  is replaced by a normalizing flow  $\hat{Z} = f(\epsilon)$ ,  $\epsilon \sim \mathcal{N}(0, I)$ , which can approximate any continuous distribution. We use a

RealNVP architecture with 6 affine coupling layers and hidden dimension 64. We additionally replace single-bandwidth MMD with a multi-scale version:

$$\overline{\text{MMD}}^2(\hat{X}, X) = \frac{1}{|\mathcal{S}|} \sum_{\sigma \in \mathcal{S}} \text{MMD}_{\sigma}^2(\hat{X}, X), \quad \mathcal{S} = \{0.5, 1.0, 2.0\},$$

and use cosine annealing on the learning rate. Critically, the Schur complement is computed on the *recovered latent representations*  $\hat{Z}^{(1)}, \hat{Z}^{(2)}$  rather than on the raw observations, as required by Theorem 1.

---

**Algorithm 3** Greedy Latent Dimension Recovery (General)

---

```

1: for  $\hat{p} = 1, 2, 3, \dots, p_{\max}$  do
2:   Initialize normalizing flow  $f_{\hat{p}}$  (RealNVP, 6 layers, hidden dim 64),  $G \leftarrow$  small random
3:   for  $t = 1, \dots, T$  do
4:     Sample  $\epsilon \in \mathbb{R}^{n \times \hat{p}}$ , rows i.i.d.  $\mathcal{N}(0, I_{\hat{p}})$ 
5:      $\hat{Z} \leftarrow f_{\hat{p}}(\epsilon)$ 
6:      $\hat{X} \leftarrow \varphi(\hat{Z})G^{\top}$ , then standardize  $\hat{X}$ 
7:      $\mathcal{L} \leftarrow \overline{\text{MMD}}^2(\hat{X}, X)$ 
8:     Update  $f_{\hat{p}}, G$  via Adam with cosine annealing
9:   end for
10:  if  $\mathcal{L} < \tau$  and  $\text{rank}(G) = n_{\varphi}$  then
11:    return  $\hat{p}, f_{\hat{p}}$ 
12:  end if
13: end for

```

---



---

**Algorithm 4** Simulation Pipeline (General)

---

**Require:**  $p, q_1, q_2, n, r, d_{\text{obs}}$ , noise family  $\mathcal{D} \in \{\text{Gaussian, Laplace, Uniform, Exponential, Mixture}\}$   
**Ensure:**  $\hat{p}, \hat{R}$

```

1: Sample weighted DAG on  $h$  nodes; partition  $Z = (Z_L, Z_{I_1}, Z_{I_2})$  with noise  $\epsilon_j \sim \mathcal{D}$ 
2: Simulate  $n$  samples from SCM
3: Generate  $X_1 = G_1(\varphi_r(Z_L, Z_{I_1}))$  and  $X_2 = G_2(\varphi_r(Z_L, Z_{I_2}))$ 
4: Form  $X_{\text{joint}} = (X_1, X_2)$ 
5:  $(\hat{d}_1, f_1) \leftarrow \text{Algorithm 3}(X_1)$ ;  $(\hat{d}_2, f_2) \leftarrow \text{Algorithm 3}(X_2)$ ;  $\hat{d}_{\text{joint}} \leftarrow \text{Algorithm 3}(X_{\text{joint}})$ 
6:  $\hat{p} \leftarrow \hat{d}_1 + \hat{d}_2 - \hat{d}_{\text{joint}}$  (Proposition 1)
7: Sample  $\hat{Z}^{(e)} \leftarrow \frac{1}{K} \sum_{k=1}^K f_e(\epsilon_k)$ ,  $\epsilon_k \sim \mathcal{N}(0, I)$ , for  $e = 1, 2$   $\triangleright K = 5$ , reduces variance
8: Compute  $\text{diag}(\hat{\Sigma}_{1|2})$  on  $\hat{Z}^{(1)}, \hat{Z}^{(2)}$  (Theorem 1)
9:  $\hat{R} \leftarrow$  indices of  $\hat{p}$  smallest entries of  $\text{diag}(\hat{\Sigma}_{1|2})$ 
10: return  $\hat{p}, \hat{R}$ 

```

---

### F.3 Hyperparameters

## References

- [1] Nils Sturma, Chandler Squires, Mathias Drton, and Caroline Uhler. Unpaired multi-domain causal representation learning. *Advances in Neural Information Processing Systems*, 36:34465–34492, 2023.
- [2] Jiaqi Zhang, Kristjan Greenewald, Chandler Squires, Akash Srivastava, Karthikeyan Shanmugam, and Caroline Uhler. Identifiability guarantees for causal disentanglement from soft interventions. *Advances in Neural Information Processing Systems*, 36:50254–50292, 2023.

Table 1: Hyperparameters used in both simulation settings.

| <b>Hyperparameter</b>   | <b>Gaussian</b>       | <b>General</b>                 |
|-------------------------|-----------------------|--------------------------------|
| Latent model            | Cholesky ( $\mu, L$ ) | RealNVP flow                   |
| Flow layers             | —                     | 6                              |
| Hidden dimension        | —                     | 64                             |
| Training iterations $T$ | 400 / 500             | 800 / 1000                     |
| MMD bandwidths $\sigma$ | {1.0}                 | {0.5, 1.0, 2.0}                |
| LR schedule             | Constant              | Cosine annealing               |
| Learning rate           | $10^{-2}$             | $10^{-2}$                      |
| Schur complement on     | $X_1, X_2$            | $\hat{Z}^{(1)}, \hat{Z}^{(2)}$ |
| Schur averaging $K$     | 1                     | 5                              |
| MMD threshold $\tau$    | 0.015                 | 0.015                          |
| $p_{\max}$              | 10                    | 10                             |
| Experiments per $n$     | 50                    | 50                             |